KiT 2.4.1 how-to

Catriona C. Conway, 2026 catriona.c.conway(at)warwick.ac.uk

Before you start

This document and the associated KiT version 2.4.1 is intended as an accompaniment for 'Single kinetochores execute an ordered series of molecular events as the Spindle Assembly Checkpoint is silenced' (Conway et al., Cell Reports, 2026), where a “MAD2 pseudo-timeline” is created to map relative ordering of events at the kinetochore over Spindle Assembly Checkpoint silencing. This document and MATLAB code should enable you to repeat the analysis yourself or utilise the pseudo-timeline for your own purposes.

If you see >> in this document, it means type whatever comes after that in the MATLAB command window, then press enter.

This pipeline has been extensively tested in MATLAB 2022a on Mac. If you have a silicon chip Mac, this pipeline will be unusable on MATLAB versions with native silicon utilisation (from 2023b onwards), so ensure you select the 'Intel' version when downloading. **The most recent MATLAB version you can use with this pipeline is 2024b.** Releases from 2025a onwards have removed the JavaFrame-based graphics rendering that we use to run KiT.

Make sure you have MATLAB (ideally 2022a) and all the following MATLAB add-ons installed (you can include these during initial MATLAB installation):

- Control System Toolbox
- Curve Fitting Toolbox
- Deep Learning Toolbox
- DSP System Toolbox
- Global Optimization Toolbox
- Image Processing Toolbox
- Instrument Control Toolbox
- Optimization Toolbox
- Parallel Computing Toolbox
- Partial Differential Equation Toolbox
- Signal Processing Toolbox
- Simulink
- Statistics and Machine Learning Toolbox
- Symbolic Math Toolbox

Download the KiT-master-2.4.1 folder from github.com/cmcb-warwick/KiT-2.4.1. Put the KiT-master-2.4.1 folder structure into your Documents / MATLAB folder.

In MATLAB, right click the KiT-master-2.4.1 folder and press Add to Path > Selected Folders and Subfolders. The folder should un-grey. If there's another KiT folder (e.g. KiD-also-known-as-KiT2.1.10-and-BEDCA-master) that isn't greyed out, right click the folder and press Remove from Path > Selected Folders and Subfolders. This is to ensure MATLAB uses the right version of KiT, which contains bug fixes for earlier versions of KiT and permitting use of all the functions described in this document (including making the pseudo-timeline graphs!).

MATLAB 2022a has a bug where closing figure windows using the ‘x’ button will crash MATLAB. The internet helped me to fix this with the creation of a startup.m file. Navigate to the 'Editor' tab on the top of the MATLAB window, then click 'New' in the top left of this tab. This will open up a new MATLAB script that you can edit. Then type:

```
set(groot, 'defaultFigureCloseRequestFcn', 'delete(gcf)');
```

Then save this in your main MATLAB folder (not the KiT subfolder) as startup.m.

In the MATLAB folder panel, make sure your main MATLAB folder is added to path by navigating to the parent folder that the MATLAB folder is sitting in. If the MATLAB folder is grey, then right click it and press Add to Path > Selected Folder.

Then in MATLAB command window type:

```
>> savepath
```

Next time you open MATLAB, this folder path will be preselected.

Things to note:

- KiT can handle a maximum of 3 channels in an image.
- Whilst this version of KiT can be used for movies, this pipeline is intended exclusively for fixed-cell images. In terms of tracking movies, KiT 2.4.1 is the same as KiT 2.4.0 (see https://github.com/cmcb-warwick/KiT).
- When you run KiT, the buttons and windows might be the wrong size to see everything on your screen. If so, you will have to edit the code so that the contents of the KiT window are visible on your screen. Please note that you will have to re-run the function to visualise any changes that you have made, if you want to make more changes then press Ctrl (not command) + C to end the function to edit the functions. You will probably need to make changes in kitSetupJob:
    - `kitSetupJob Line 53: colw = [55 50];`
    - This is how big the left and right columns are in the setup window (in characters). Increment each by a small amount.
    - `kitSetupJob Line 55: figh = 53;`
    - This is how tall the setup window is. Increment by 1 or 2.
    - There may be other various buttons and spacings that you may have to change to fit the window/your screen in kitSetupJob – I’m afraid you will have to dig in the code to find the relevant variables to change. The titles of each tab/subsection are commented above the section of code corresponding to creating and laying out that tab/subsection, so that should help you to locate the appropriate regions to change.
- KiT has been written in British English. If you are getting errors with the options, you may be using a version of a word that contains a ‘z’ rather than an ‘s’, or needs a ‘u’!

1 Chromatic shift

This data should come from a set of ~10 cells (ideally all mitotic, but can include anaphase) which have the same kinetochore marker labelled in the same colours as your experimental images, with images collected on the same day as your experimental images. McAinsh lab generally uses wild-type RPE1 cells with CenpC (MBL PD030) as a primary antibody and then secondaries are a single mixture of 3 anti-guinea pig antibodies conjugated with 1) AlexaFluor 488, 2) AlexaFluor 594, 3) AlexaFluor 647. We image with 100 nm z-spacing to enable fine chromatic shift correction. <u>You can use a single chromatic shift dataset for all experiments imaged on the same day.</u>

Make sure you have provided your chromatic shift images with the same channels in the same order as your experimental images, e.g. if channels 1, 2, and 3 respectively correspond to far-red, red, and green for your experimental data, then the same should be true for your chromatic shift data. If you have imaged 3 channels for chromatic shift, but only 2 channels for your experimental data, go into Fiji and remove the extra channel in the chromatic shift images. I've never tried to analyse with anything other than the same channels in the same order for experimental and chromatic shift datasets, but I'm sure KiT will complain and/or give you incorrect outputs if you do try this.

1.1 Set up and run chromatic shift job

MATLAB command window

```
>> kitRun
```

KiT 2.4.1 window

Press <u>New job</u>

Select <u>Chromatic shift</u>

Job setup window

Section: *2. Process setup*

<u>Detect spots in channel…</u> select the channel which contains your kinetochore reference marker in your experiment dataset. **IMPORTANT:** channels 1, 2, and 3 represent the order of channels in the images that you have provided to KiT. E.g. if you have a two-channel image where channel 1 is far-red and channel 2 is green, then you will only be able to select channel 1 or channel 2, channel 3 will be greyed out. The channels are kept in the same order as provided to KiT, i.e. the same order that your channels would be shown in Fiji. KiT tends to assume your images are provided in green, red, far-red order, so you may have to change KiT's downstream assumptions to match your data.

<u>Spot detection…</u> keep Histogram selected

<u>Spot refinement…</u> keep MMF selected

<u>Use channel (1/2/3) to detect spots in channels…</u> tick other box(es).

Section: *3. Options*

Tab: Detection

Max spots per frame, choose a number that is the maximum number of kinetochores you expect for your cell line plus ~50 (this is to give KiT the opportunity to find dim kinetochores whilst still excluding most background signal)

Tab: MMF

Tick using deconvolved data if this is the case (this is true for McAinsh lab data).

Weight for intensity restriction in channel… If your images are not 3 channel and/or not supplied to KiT in green > red > far-red order, make sure that you use the appropriate number in each channel. This is 0.05 for green, 0.075 for red, and 0.1 for far-red.

Section: *1. Movie and ROI selection*

Press Select directory

Navigate to folder with Chromatic Shift images, press open

Make sure it's only images selected, not .txt files!

Selected movies: Add and crop

Press Add ROI

Draw box around cell, including some background

Double click in box so it turns yellow

Press finish

Do for all cells

Section: *4. Final checks*

Give it a file name (like exp01_ChromaticShift)

Press Validate metadata

Metadata window

Assuming all of the metadata is the same for each image (i.e. wavelengths and channel order, xyz dimensions, pixel size), ensure that all of the metadata is as you expect and the wavelengths are the correct order for your channels, then:

Tick apply to all movies

Press Validate

Otherwise, if some of the metadata is different for some images, you will have to check and validate all metadata individually for each image.

Job setup window

Press Save jobset

KiT 2.4.1 window

Press Run

Wait ~5-10 mins, command window will say 'Tracking complete'

1.2 Filter spots

KiT 2.4.1 window

In **section 2** press Spot filtering.

For each of your channels: if any of the following are true, remove spot:

- Spot centre is poorly detected (i.e. central cross is not in middle of spot)
- Spot is misshapen, i.e. is not Gaussian and symmetrical (mildly elliptical is okay).
- Spot is close to another spot (use intensity circle option in the bottom to check this, if the edge of the intensity circle is touching another spot, remove).
- Spot is very faint.
- Spot isn't a kinetochore from your chosen cell. If you press change zoom, then you'll see the whole cell (if you press it again, you'll zoom back in). This will permit you to see if a spot is a kinetochore (i.e. is within the mass of other spots) or is background noise (generally far away from other spots or near the edge of the cropped region).

Press Next Chan (bottom) to go through all channels (the name of the window will change to reflect which channel you are currently assessing), then Next (top right) to go through all cells. Press Finish once all cells have been filtered.

2 Spot detection on experimental data

You should do a new set of spot detections with new kitjobset name for each experiment and experimental condition (e.g. if you have exp01 with DMSO and nocodazole treatment, you should perform this section twice, splitting by treatment – this makes downstream analysis much easier). For the MAD2 pseudo-timeline, we performed this analysis on cells between early prometaphase and metaphase (inclusive). We did 200 nm z-spacing as we found that balanced image size/imaging time with good spot centre localisation.

MATLAB command window (if you closed the KiT window after section 1)

```
>> kitRun
```

KiT 2.4.1 window

Press New job

Select Single timepoint

Job setup window

Section: *2. Process setup*

Coordinate system… keep Centre of mass selected

Detect spots in channel… select the channel that you imaged your reference kinetochore marker in. **IMPORTANT:** channels 1, 2, and 3 represent the order of channels in the images that you have provided to KiT. E.g. if you have a two-channel image where channel 1 is far-red and channel 2 is green, then you will only be able to select channel 1 or channel 2, channel 3 will be greyed out. The channels are kept in the same order as provided to KiT, i.e. the same order that your channels would be shown in Fiji. KiT tends to assume your images are provided in green, red, far-red order, so you may have to change KiT's downstream assumptions to match your data.

Spot detection… keep Histogram selected

Spot refinement… keep MMF selected

Use channel (1/2/3) to detect spots in channels… tick other box(es). This gives greater downstream intensity measurement functionality.

Section: *3. Options*

Tab: Detection

Channel orientation, use buttons to change to your predicted inner/outer sequencing of kinetochore markers. This ordering will be used later for accessing intensity measurements and correcting fine chromatic shift aberrations so it's very important you change this to match your experimental setup!

Max spots per frame, choose a number that is the maximum number of kinetochores you expect for your cell line plus ~50 (this is to give KiT the opportunity to find dim kinetochores whilst still excluding most background signal)

Tab: MMF

If appropriate (is for McAinsh lab), tick using deconvolved data

Weight for intensity restriction in channel… If your images are not 3 channel and/or not supplied to KiT in green > red > far-red order, make sure that you use the appropriate number in each channel. This is 0.05 for green, 0.075 for red, and 0.1 for far-red.

Tab: Chromatic shift

Tick provide chromatic shift correction

Tick filter spots. Don't change the parameters of filtering unless your data dictates it.

For each box with '-' in it, select box, navigate to relevant chromatic shift kitjobset file (e.g. 'kitjobset_231025_exp01_ChromaticShift.mat', not the kittracking file), press open

Tab: Intensity

Measure in channels… Tick all channels

Tab: Tasks

Make sure everything is ticked

Section: *1. Movie and ROI selection*

Press Select directory

Navigate to folder with experiment images, press open

In the job setup window, make sure only images are selected, not .txt files!

Selected movies: Add and crop

Press Add ROI

Draw box around mitotic cell (include some background. Include spindle poles if visible)

Double click in box so it turns yellow

Press finish

Do for all cells

Section: *4. Final checks*

Give it a name. Something descriptive like exp01_Mad2CenpCSKAP (experiment number, epitopes in the channel order you have provided them to KiT)

Press Validate metadata

Metadata window

Assuming all of the metadata is the same for each image (i.e. wavelengths and channel order, xyz dimensions, pixel size), ensure that all of the metadata is as you expect and the wavelengths are the correct order for your channels, then:

Tick apply to all movies

Press Validate

Otherwise, if some of the metadata is different for some images, you will have to check and validate all metadata individually for each image.

Job setup window

Press Save kitjobset

KiT 2.4.1 window

Press Run

Wait about 1.5h for 80 cells! Command window will say 'Tracking complete' once finished. This will create an individual kittracking file for each cell, and a kitjobset file which is a pointing file for the whole experiment and contains information on where the individual kittracking files are, what the cropping dimensions are, etc.

Close the KiT window.

Checking data quality (optional)

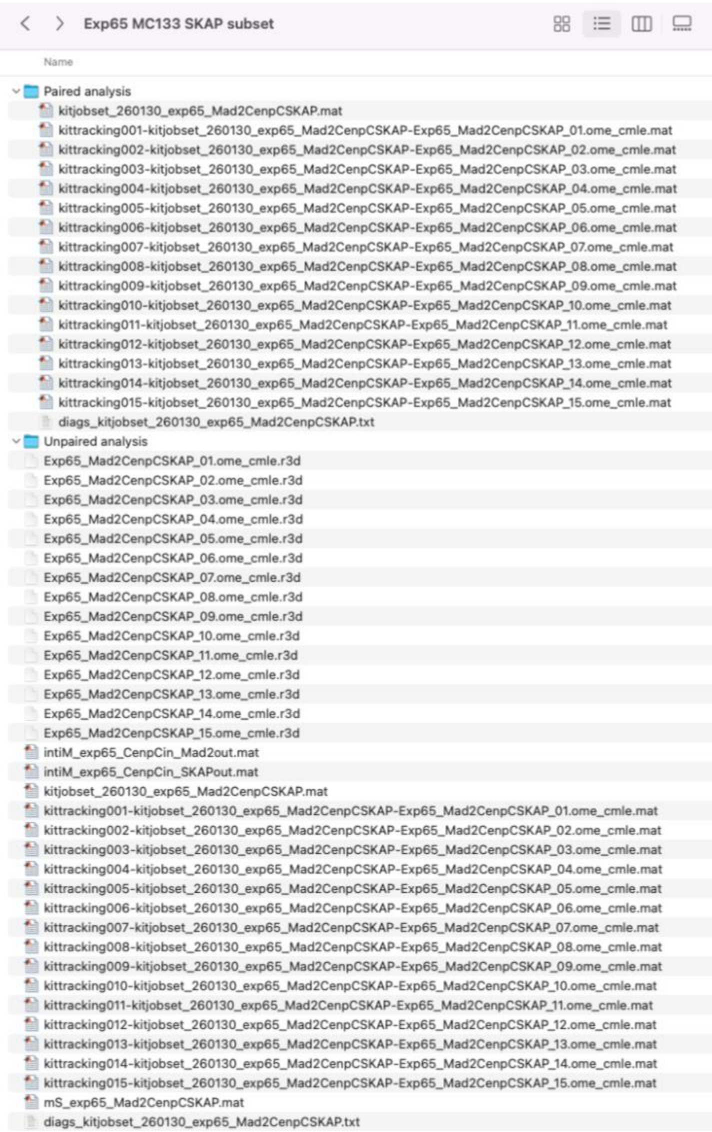
To see how well spot detection has worked, navigate to the folder with your images and open diags_yourjobsetname.txt. In the second column you'll see how many spots have been detected in each cell. We want at least 46 spots per cell (but >80 is ideal). Sometimes a cell will fail detection and that is okay, it will be automatically skipped in downstream analysis.

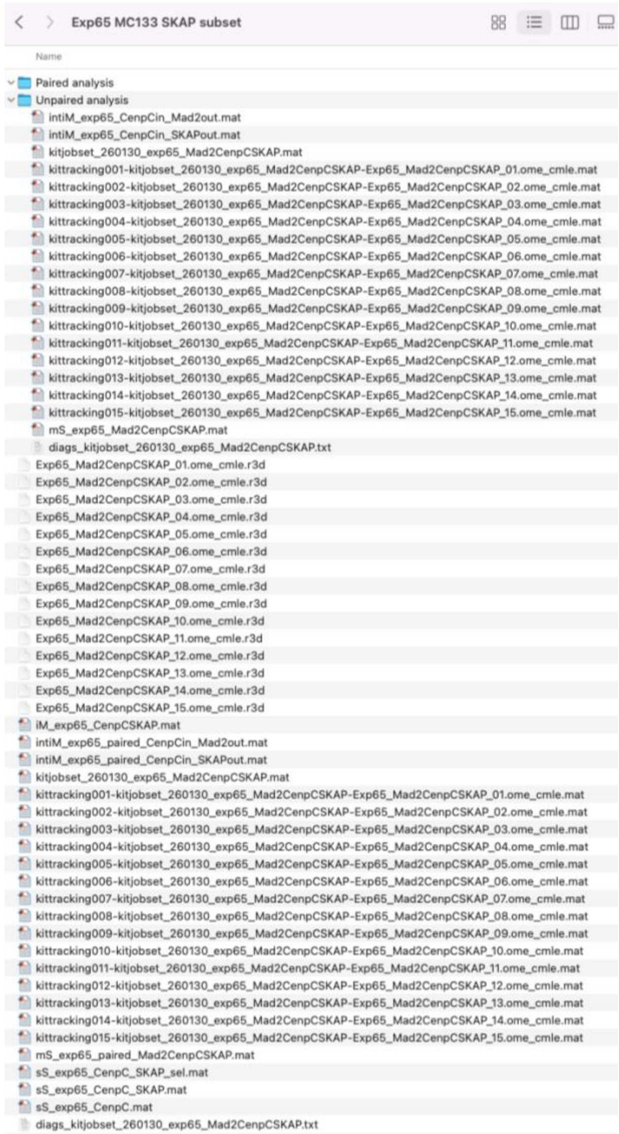
3 Deciding on data analysis for pseudo-timeline plotting

There are three things that you may want to plot on the pseudo-timeline: Kinetochore intensity measurements, inter-sister distances, and intra-kinetochore Delta distances. These all follow a slightly different analysis pipeline from each other.

Considerations:

- Are you just assessing intensity **OR** intra-kinetochore Delta and/or inter-sister distance (and don't care if you're not using the same kinetochores for intra-kinetochore Delta and inter-sister distance)? If so, you can get away with one round of tracking with no messing about with folders.
  - See section 4 for 3-channel intensity analysis
  - See section 5 for inter-sister distance measurements
  - See section 6 for intra-kinetochore Delta measurements
- Are you assessing three channel intensity **AND** intra-kinetochore Delta and/or inter-sister distance? Do you want to get as much data as you can for each individual graph for your experiment (i.e. pairing for all measurements isn't something you're worried about)? If so, you should duplicate your kitjobset and kittracking files, and put the new copy into a 'pairing' subfolder – this is because spot filtering for intensity measurements messes up pairing, and vice versa. I advise that you do intensity measurements first (as it takes the shortest), then do inter-sister distance measurements, the initial analysis of which you can then use to do intra-kinetochore Delta distance measurements as your final step.
  - During 3-channel intensity analysis (see section 4), have the unpaired kitjobset/kittracking files in the same directory as your image files, and the kitjobset/kittracking files to be paired in a separate subfolder.
  - During inter-kinetochore distance measurements (see section 5) and intra-kinetochore Delta measurements (see section 6), have the kitjobset/ kittracking files which need to be/are paired in the same directory as your image files, and the kitjobset/kittracking files from the 3-channel intensity analysis in a separate subfolder. See next page for a side-by-side comparison.
- Do you want to plot data from exactly the same kinetochores in multiple different ways? If so, then you should do pairing and spot selection first.
  - If you wish to perform all of 3-channel intensity analysis, inter-sister distance measurements, and intra-kinetochore Delta measurements, follow section 7.
  - If you wish to perform 3-channel intensity analysis and intra-kinetochore Delta measurements, follow section 7 except 7.6.
  - If you wish to perform inter-sister distance measurements and intra-kinetochore Delta measurements, follow section 7 except 7.7 and 7.8.
  - If you wish to perform 3-channel intensity analysis and inter-sister distance measurements, follow section 8.

<table>
<tr>
<td>Placing of kitjobset/kittracking files during unpaired intensity measurement analysis (section 4) – note that the 'Unpaired analysis' folder is empty</td>
<td>Placing of kitjobset/kittracking files during paired inter-kinetochore distance and intra-kinetochore Delta analysis (sections 5 and 6) – note that the 'Paired analysis' folder is empty</td>
</tr>
<tr>
<td>


</td>
<td>


</td>
</tr>
</table>

4 Analysing three-channel intensity data

4.1 Spot filtering on unpaired data

You can choose to do this through the KiT 2.4.1 main window, but I prefer to do it through the command window as it gives you a couple more options.

IMPORTANT: if you close MATLAB, you lose the workspace variables!!! Make sure to save them by right-clicking them in the workspace and saving.

4.1.1 Load the kitjobset file into the workspace

You can't just drag in your desired kitjobset from its saved location, sadly. You have to do it through a KiT function.

Your kitjobset name begins with 'kitjobset_YYMMDD_'. Whatever comes after that is the name you input in section 2. I will be referring to that as 'name' for the rest of this document.

Type in the MATLAB command window:

```
>> kitjobset_YYMMDD_name = kitLoadJobset;
```

This will open up file explorer. Navigate to the experimental data kitjobset file you've just created (e.g. 'kitjobset_231025_exp01_Mad2CenpCSKAP.mat', always choose the kitjobset rather than kittracking file) and press open. This will load in your kitjobset file to your MATLAB workspace. If you load in the kittracking file, this will only consider one cell and your code will also throw an error later down the line.

Don't put .mat at the end of your kitjobset name!

4.1.2 Filter the spots

Now we can filter the spots:

```
>> kitFilterSpots(kitjobset_231025_exp01_Mad2CenpCSKAP, 'suggestExcl', 1);
```

'suggestExcl' will auto-exclude spots which are below a dimness threshold (you can disagree with KiT's assessment and include these spots if you like).

Look at the spot in the reference kinetochore marker channel only, and if any of the following are true, remove:

- Spot centre is poorly detected (i.e. central cross is not in middle of spot)
- Spot is misshapen, i.e. is not Gaussian and symmetrical (mildly elliptical is okay)
- Spot is close to another spot (use intensity circle option in the bottom to check this, if the edge of the intensity circle is touching another spot, remove).
- Spot is very faint
- Spot isn't a kinetochore or isn't from your chosen cell. If you press change zoom, then you'll see the whole cell (if you press it again, you'll zoom back in). This will permit you to see if a spot is a kinetochore (i.e. is within the mass of other spots) or is

background noise (generally far away from other spots or near the edge of the cropped region).

Press next after each cell. Once you've done all cells, press finish and a small amount of tracking will re-run (but it won't take 1.5h!).

Optional: If you're not done but you have to stop the selection for a substantial amount of time (e.g. when it's the end of the day, you have another experiment to do, etc), just press finish and let it re-run. Make a note of the next cell you need to filter from. Then when you come back to it, type the following, and then follow the process again. (XX should be the number of the next cell you need to filter from).

```
>> kitFilterSpots(kitjobset_231025_exp01_Mad2CenpCSKAP, 'suggestExcl', 1,
'startMovie', XX);
```

If MATLAB crashes and you are unsure which cell you have filtered up to, navigate to the folder that contains your kittracking files and check when they were last edited. The most recently edited file will have successfully recorded its filtering. When you re-run the filtering function, the ‘startMovie’ should be the cell after the last one that recorded its filtering.

4.2 Creating the intensity measurement files

4.2.1 Create a movieStructure file

This will distil all the information in the kittracking and kitjobset files into one structure that you can then navigate through in MATLAB.

In the MATLAB window, type the following:

```
>> mS_name = kitLoadAllJobs(kitjobset_YYMMDD_name);
```

Example:

```
>> mS_exp01_Mad2CenpCSKAP =
kitLoadAllJobs(kitjobset_231025_exp01_Mad2CenpCSKAP);
```

Copy the mS name, right click it in the workspace and click save as, navigate to the folder you wish to save in, paste the mS name (right click to paste), then save. You can drag this back into the workspace to reload it once you've closed MATLAB.

4.2.2 Create intensity structures

This will extract intensity values from the movieStructure for the channels you select (can only be 2 channels at a time). Points to consider:

- This defaults to accessing the intensity measurement in the vicinity of the 'inner' spot (with position corrected for chromatic shift for the secondary channel). We are assuming you have designated the reference kinetochore spot channel as the innermost channel back in Section 2.
    - If you aren't sure on what order your channels are in, open up your movieStructure within MATLAB, then navigate to cell 1 > options > neighbourSpots > channelOrientation. This will list your channels from inner to

outer. You can also access this in the command window (no semicolon will permit the output to be printed to the command window):

```
>> mS_name{1}.options.neighbourSpots.channelOrientation
```

- If you do not want the intensity to be measured in this way, change 'refMarker' to 'outer' for the intensity to be measured in the vicinity of the marker you designated as being 'outside' in Section 2, or 'self' for each intensity to be measured in the vicinity of the detected spot for their own channel (if detected, not recommended in the case of variable epitopes like Mad2).
- If you have your 'outer' channel as the reference channel, that makes downstream graph plotting awkward as I based literally all of my plotting scripts on the reference kinetochore marker being ‘inner’. Swap the inner and outer intensity measurements as shown in highlighted region below.

- Ideally you would normalise to your reference kinetochore marker, so 'channels' should include the reference kinetochore marker channel and the channel number of your intensity protein.

If you want to access intensity measurements for all three channels, you should create two separate intensity structures:

```
>> intiM_exp01_CenpCin_Mad2out = dublIntensityMeasurements(mS_name, 'channels', [1 2])

>> intiM_exp01_CenpCin_SKAPout = dublIntensityMeasurements(mS_name, 'channels', [2 3])
```

This is because you will probably want to measure the intensity of the proteins in the same (chromatic shift-corrected) location as one another. This is most reliable when you are using the spots detected in the reference kinetochore marker channel, as they should be present throughout mitosis and not subject to substantial changes that may affect their detection.

These intiMs are named assuming that these proteins are indeed in this order – verify this as described above.

If you want to swap the orientation of your channels (heavily advised for downstream plotting if your reference channel is in the 'outer' position), please use the following function:

```
>> intiM_exp01_CenpCin_Mad2out = kitSwapIntiMInnerOuter(intiM_exp01_CenpCout_Mad2in);
```

Save your intiM structures in the same way as the movieStructure. You can drag these back into the workspace to reload them once you've closed MATLAB.

4.3 Plotting intensity measurement pseudo-timeline graphs

The following function will split your kinetochores into bins based on their Mad2 intensity, then plot the intensity of Mad2 (in black) and your second marker (in pink) within those bins.

```
>> intData = makeIntPlots(ctrlIntiMs, treatIntiMs)
```

ctrlIntiMs should contain intiMs from untreated, DMSO-treated, and/or wild-type cells (matched with treatIntiMs, if appropriate). They can be from a single experiment or multiple experiments. They should be in curly brackets, with the Mad2 intiM first and the additional marker intiM second, separated by commas. If you have multiple experiments, each experiment should be separated by a semicolon, with commas in between the intiMs for a single experiment. For example:

```
{intiM_exp01_CenpCin_Mad2out, intiM_exp01_CenpCin_SKAPout;
intiM_exp02_CenpCin_Mad2out, intiM_exp02_CenpCin_SKAPout}
```

If you have multiple experiments (as shown here), there will be a first-pass ordering of kinetochores and normalising to 1 for each experiment; the normalised data from all experiments will then be combined, re-sorted and re-binned.

treatIntiMs should contain intiMs from drug-treated/perturbed cells, (e.g. knockout/ knockdown cell lines, or treated with nocodazole / MG132 / ZM / reversine / monastrol etc) matched with their corresponding ctrlIntiMs. **If you only have non-perturbed data, input this variable as [ ] (empty square brackets).** This function will normalise the treatIntiMs to the ctrlIntiM in the same position as it, e.g. if you had the following treatIntiMs:

```
{intiM_exp01_ZM_CenpCin_Mad2out, intiM_exp01_ZM_CenpCin_SKAPout;
intiM_exp02_ZM_CenpCin_Mad2out, intiM_exp02_ZM_CenpCin_SKAPout}
```

Your ctrlintiMs should be ordered as such:

```
{intiM_exp01_DMSO_CenpCin_Mad2out, intiM_exp01_DMSO_CenpCin_SKAPout;
intiM_exp02_DMSO_CenpCin_Mad2out, intiM_exp02_DMSO_CenpCin_SKAPout}
```

This will normalise your treatment intensities to your control intensities (matched for each experiment and marker), split into the same number of Mad2 bins as the control, and plot.

YOU CANNOT HAVE TREATINTIMS WITHOUT CTRLINTIMS !! I've never tried it but I'm 10000% sure attempting it will throw you an error.

There are options that you can add after treatIntiMs:

- 'nBins', 25 is default but you can change to any integer
- 'normIndInt' is 1 by default, but can change to 0. This normalises the Mad2 intensity to the reference kinetochore marker intensity (assuming these are arranged as 'outer' and 'inner', respectively)
- 'normDepInt' is 1 by default, but can change to 0. This normalises the intensity of your variable protein to the reference kinetochore marker intensity (assuming these are arranged as 'outer' and 'inner', respectively)
- 'normalise' is 1 by default, but can change to 0. This scales all intensities such that the maximum median intensity for each marker is 1 for the control graph. I would advise keeping this as 1 if you have multiple experimental repeats.
- 'indMarker' is 'Venus-MAD2' by default but you can change this to any string. This is the epitope that you ordering the kinetochores by for binning.

- 'depMarker' is empty by default but you can change this to any string. This is the secondary marker whose intensity you are assessing after binning.
- 'normMarker' is 'CENP-C' by default but you can change this to any string. This is the marker that you are normalising by (generally the reference kinetochore marker). If you are not normalising to anything, leave this as-is because it won't be used.

Example: plotting one experiment without treatment data in 20 bins

```
>> exp01_Mad2ind_SKAPdep = makeIntPlots({intiM_exp01_DMSO_CenpCin_Mad2out,
intiM_exp01_DMSO_CenpCin_SKAPout}, [], 'nBins', 20, 'indMarker', 'Mad2',
'depMarker', 'SKAP');
```

Example: plotting two experiments including treatment data in 25 bins

```
>> exp01_02_DMSO_ZM_Mad2ind_SKAPdep =
makeIntPlots({intiM_exp01_DMSO_CenpCin_Mad2out,
intiM_exp01_DMSO_CenpCin_SKAPout; intiM_exp02_DMSO_CenpCin_Mad2out,
intiM_exp02_DMSO_CenpCin_SKAPout}, {intiM_exp01_ZM_CenpCin_Mad2out,
intiM_exp01_ZM_CenpCin_SKAPout; intiM_exp02_ZM_CenpCin_Mad2out,
intiM_exp02_ZM_CenpCin_SKAPout}, 'indMarker', 'Mad2', 'depMarker', 'SKAP');
```

The created variable intData will allow you to access the normalised data that created the graph. The 'independent' portion is the marker that the data has been binned based on (generally Mad2, the first entry of your intiMs) and the 'dependent' portion is the data from the secondary marker (the second entry from your intiMs). If you have treatment and control datasets, the values for both of these can be accessed. The code will automatically calculate 'half change' for Mad2 and your secondary marker, i.e. the bin where the intensity value of each of these has changed by half, and the intensity of Mad2 at the location where the intensity value of your secondary marker has changed by half. Save intData to easily access the values used for plotting these graphs later on.

Your intensity curves will have three regions – a light shaded region corresponding to the lower and upper quartiles, a dark shaded region corresponding to the 95% confidence interval of the median, and a solid line corresponding to the median.

You can alter the dimensions of the plot relatively easily through the MATLAB command window (I recommend changing axis limits and dimensions in MATLAB as it's a pain to do neatly in vector graphics software). I have made the code so that the treatment data axes are at the same scale as the control data – unfortunately, this might make the axes larger than can fit in your figure window/screen! If this happens, I recommend selecting your too-big figure, then typing the following in the MATLAB command window:

```
>> gca_pos = get(gca, 'Position')
```

This will print (in order) the left position, bottom position, width and height dimensions of your axes in pixels. You can then type:

```
>> get(0, 'ScreenSize')
```

This will give you your screen 'position' in pixels in the same order as the axes 'position' above. This will give you an idea of the relative scales that you are working with. Then you can change your axes 'Position' to work within that. You probably want to leave your left/bottom/width positions the same.

```
>> set(gca, 'Position', [gca_pos(1), gca_pos(2), gca_pos(3), NEW HEIGHT])
```

If you want to change the y-limits of the left (Mad2) or right (other protein) y-axis, then type the following, altered for what you want to change and what you want to change to:

```
>> yyaxis left
```

```
>> ylim([-0.05, 1.2])
```

This will keep the size of your axes the same, but will stretch/compress your data within that.

Once you are happy with your plot, in figure window, File > Save as, then save as .fig file. This will only open in MATLAB, but is best to access and use if you need to change axis scaling etc.

You can also save as .svg file for editing in vector programmes – you may have to change the figure renderer to 'painters' in MATLAB for this to be editable in your vector editing software (it may just export as a picture in .svg format otherwise). You can do this through the command window:

```
>> set(gcf, 'Renderer', 'painters')
```

5 Inter-sister distance measurements

For when you want to measure the inter-sister distances between pairs of kinetochores. Even though you can plot inter-sister distances using section 6, I advise that you use this section if you have not assigned the channel containing your reference kinetochore marker as the innermost channel.

5.1 Pairing

Unambiguously the worst bit of this analysis. You can realistically pair 30-40 cells a day and you will also be bored out of your skull. I would recommend having a TV show on in the background, ideally one you've watched before! You can use the KiT 2.4.1 window pairing, but I personally prefer to do this through the code as the code-based pairing gives you more flexibility about quitting and restarting (which you will definitely want to do).

5.1.1 Load kitjobset

If you have done unpaired 3-channel intensity measurements, remember to move your unpaired kitjobset/kittracking files used in intensity measurements to their storage folder, and move the unedited kitjobset/kittracking files for pairing to the same folder as the images (KiT will throw an error otherwise as the directory of the kitjobset/kittracking files will not match what they are supposed to be).

If you forgot to duplicate your kitjobset/kittracking files and have already done intensity measurements, make a new subfolder called 'Analysed unpaired' or similar and place a copy of your analysed kitjobset/kittracking/movieStructure/intiM files into there. **If you don't duplicate and just continue anyway, the raw files for your original analysis will be overwritten.** Load in the kitjobset file from the folder containing your images as below.

Load in the kitjobset file from the folder containing your images using the KiT function below.

Your kitjobset name begins with ‘kitjobset_YYMMDD_’. Whatever comes after that is the name you input in part 2. I will be referring to that as ‘name’ for the rest of this document.

Type in the MATLAB command window:

```
>> kitjobset_YYMMDD_name_paired = kitLoadJobset;
```

This will open up file explorer. Navigate to the non-analysed or freshly duplicated kitjobset file which is in the same folder as your images (always choose the kitjobset rather than kittracking file, e.g. 'kitjobset_231025_exp01_Mad2CenpCSKAP.mat') and press open. This will load in your kitjobset file to your MATLAB workspace. If you load in the kittracking file, this will only consider one cell and your code will also throw an error later down the line.

Don't put .mat at the end of your kitjobset name!

If you loaded in a duplicated, analysed kitjobset, you can now use a function to revert the data to the raw data.

```
>> kitRevertToRawData(kitjobset_YYMMDD_name_paired)
```

You can now continue as normal.

5.1.2 Pair

You have options! You have to put these options in single quotes and then what you want from these options afterwards. Its default options are:

- 'mode', 'dual' (a zoomed and full cell XY image)
    - 'mode', 'triple' will give you an additional XZ projection (basically you are looking at the cell from the bottom edge of the image). I like this as early prometaphase cells can have their kinetochores arranged vertically and sometimes the metaphase plate is slanted. This gives you a 3D idea of where your kinetochore and its pairing candidates are, but I know it can be a confusing perspective so I won't be offended if you don't use it.
- 'maxSisSep', 2.2 (in microns – you can change this, but I never go less than 2 or more than 2.5)
- 'autoSize', 1 (change to 0 if you do not want it to automatically maximise to your main window)
- 'subset', [] (change to [startingCell:finalCell] if you stopped part way through pairing and want to pick up where you left off)
- 'redo', 1 (I would keep this as it is, it lets you re-pair an image)
- 'imageChans' and 'plotChan', 2 (change to another channel if you want to pair in that channel, e.g. if you have your kinetochore reference marker in channel 3, change both of these options to 3)
- 'zProjRange', 2 (max intensity projection of 2 slices above and below spot plane, therefore you see max intensity projection of 5 slices – I wouldn't change this)

Example – you want to see xy and xz projections, and your reference kinetochore marker is in channel 3, and you want the pairing window to automatically maximise:

```
>> dublManualPairSisters(kitjobset_YYMMDD_name_paired, 'mode', 'triple',
'autoSize', 1, 'imageChans', 3, 'plotChan', 3);
```

Now you get the first cell opening up, press any key except 'n' or 'q' to pair that cell. ('n' will skip that cell, and 'q' will quit if you want to come back to it later – see later for how to do this).

If you are using a mac (untested on Windows), on your first time running this, it will request that you give permission for 'createDistanceMatrix.mexa64' (or maybe another file extension based on your system) to open. **Do not move it to bin!** Click allow and input your password. This will almost certainly crash the pairing, so after you have allowed the function to work, press Ctrl + C (not command!) to cancel this round of pairing, then close the figure window for pairing, and call the dublManualPairSisters function again (note that pressing the 'up' arrow on your keyboard whilst in the MATLAB command window lets you access recently called functions). You should now be able to pair, and you will probably never have to specifically allow this file to be accessed again.

If there is a metaphase plate visible, utilise that to guide your pairing (generally perpendicular to the metaphase plate). Keep in mind that kinetochores in earlier stages of mitosis are more likely to not be following the metaphase plate, and could be oriented in a rosette shape (where one sister is closer to the centre and the other is outward). You may also have kinetochore pairs where sisters aren't visible in xy but are relatively close together in the z dimension (this is why I added in the 'triple' view functionality). Deciding which kinetochores are pairs is a judgement call that you have to make based on your experience and general vibes of the cell/kinetochores, especially in early prometaphase.

You start with kinetochore 1 and go until there are no kinetochores left unpaired or unskipped. Press space to skip a kinetochore if it is obviously bad (see 4.1.2 for criteria for this). Otherwise, click the kinetochore you think is the pair of the kinetochore under consideration. Unpaired kinetochores are in green, skipped kinetochores are in yellow, and paired kinetochores are in red. I would encourage you to choose the kinetochore you think would be the best pair for your kinetochore, irrespective of whether it's already been paired or not. Pairing to an already-paired kinetochore will just remove its previous pairing. Press space to skip a kinetochore if a) there are no realistic candidates for it to be paired with (or the realistic candidate hasn't been detected for some reason – this happens sometimes), or b) if there are 2 kinetochores that could pair with the same central kinetochore and you're just going back and forth between them – choose the best kinetochore for the pair and skip the other one. If you have a really nice kinetochore but you think its pair is a really bad one, just pair the kinetochores and you can then filter out the bad kinetochores later.

If you want to pause for a while and re-start later, press 'q' when KiT asks if you want to continue pairing the next cell, then make a note of the next cell you need to pair. Then, when you want to come back to pairing, put the following into the command window. Here, firstCell is the cell you want to repeat/continue from and lastCell is the number of cells you have:

```
>> dublManualPairSisters(kitjobset_YYMMDD_name_paired, 'mode', 'triple',
'autoSize', 1, 'subset', [firstCell:lastCell]);
```

If you've messed up a cell (e.g. you have been pairing believing the metaphase plate being in a different orientation than it actually is), you have two options: Ctrl + C (not command!) will exit the pairing immediately, or you can wait until the cell has finished, and then press 'q' on the next cell. Then use the above command, with firstCell being the erroneous cell.

If MATLAB crashes and you are unsure which cell you have paired up to, navigate to the folder that contains your kittracking files and check when they were last edited. The most recently edited file will have successfully recorded pairing. When you re-run the pairing function, firstCell in 'subset' should be the cell after the last one that recorded pairing.

5.1.3 Create a movieStructure file

This will distil all the information in the kittracking and kitjobset files into one structure that you can then navigate through in MATLAB.

In the MATLAB window, type the following:

```
>> mS_name_paired = kitLoadAllJobs(kitjobset_YYMMDD_name_paired);
```

Example:

```
>> mS_exp01_paired_Mad2CenpCSKAP =
kitLoadAllJobs(kitjobset_231025_exp01_paired_Mad2CenpCSKAP);
```

Copy the mS name, right click it in workspace and click save as, navigate, paste the mS name (right click to paste), then save.

5.2 Spot selection

5.2.1 If you have already selected good quality reference kinetochore marker spots for intensity measurements

If you have already screened your reference kinetochore marker spots for intensity measurements in 4.1.2, you can save some time by using the screened data to make spot selections for your new paired data via the following function:

```
>> sS_name_CenpC = kitGetPairedSpotSelFromFilteredData(mS_unpairedFiltered,
mS_pairedUnfiltered);
```

Here, mS_unpairedFiltered is the structure you obtained in 4.2.1 and mS_pairedUnfiltered is the structure you obtained in 5.1.3.

This will obtain the IDs of the acceptable spots from 4.1.2 and compare them to the spots you have accepted for pairing, and then remove any spots which did not pass your screening during 4.1.2.

Save this sS structure. Continue to 5.3.

5.2.2 If you have not done any spot filtering for this experiment at any point

You may want to edit the kitSelectData_sS file. Because of past errors that have made me lose hours of work, I have instructed the function to save the spot selection as it goes in a folder, which is currently the folder you have open in the MATLAB panel. If that is suboptimal for you, e.g. if you have an analysis folder somewhere else you want to save to, then replace kitSelectData_sS line 93 with the following:

```
filename = sprintf('FILEPATH_TO_DESIRED_FOLDER%s%s_workingSpotSelection.mat',
folderSeparator, todayDate);
```

If this is giving you errors, make sure there are no extra forward/back slashes at the end of ‘Filepath to desired folder’.

Do a spot selection **only** for your kinetochore reference marker:

```
>> sS_name_CenpC_sel = kitSelectData_sS(mS_name_paired);
```

This will open up a window asking if you want to continue for each cell. Click 'yes' if you want to continue, 'skip' if you don't want to use this cell for some reason, or 'no' if you want to pause for a bit. Remember to save your sS structure if you’re pausing.

For your reference kinetochore marker channel only, look at the spot, and if any of the following are true, remove:

- Spot centre is poorly detected (i.e. central cross is not in middle of spot)
- Spot is misshapen, i.e. is not Gaussian and symmetrical (mildly elliptical is okay)
- Spot is close to another spot (use intensity circle option in the bottom to check this, if the edge of the intensity circle is touching another spot, remove).
- Spot is very faint

Don't click 'next chan' as that will result in you filtering in more than the reference kinetochore marker channel, you may be then forced to filter out good kinetochores because of lack of detection of a spot in other channels (which may be expected if any of your other proteins are expected to be absent from the kinetochore at any point).

Press finish after each cell. Once you've done all cells, you will get a spot selection structure.

Optional: If you're not done but you have to stop the selection for a substantial amount of time (e.g. when it's the end of the day, you have another experiment to do, etc), just press "no" when asking to continue spot selection. Make sure you save the spot selection so far. Then when you come back to it, take your old sS_name file and do 'addMore', sS_name as an option, and 'startMovie' as the next cell you need to start selecting from, e.g. if you completed selection up to cell 20 and now want to start from cell 21:

```
>> sS_name_CenpC_sel2 = kitSelectData_sS(mS_name_paired, 'addMore',
sS_name_CenpC_sel, 'startMovie', 21);
```

If you have to pause multiple times, then change the variable names to reflect how many times you've had to pause, and make sure your startMovie number is correct, e.g.:

```
>> sS_name_CenpC_sel3 = kitSelectData_sS(mS_name_paired, 'addMore',
sS_name_CenpC_sel2, 'startMovie', 42);
```

Increment each sel number by 1 each time you re-start for the same experiment's spot selection. Save your sS structure every time you pause.

If MATLAB crashes, navigate to the folder that contains your 'workingSpotSelection', and drag it back into MATLAB. If you are unsure which cell you have got to, click into that spot selection, then navigate to the data-containing portion of the 'selection' substructure and scroll to the bottom. The number at the bottom of the left-hand column is the last cell that you performed spot selections for. You can then continue spot selections as described in the highlighted blue section above, using the workingspotselection structure as the selection for 'addMore' in place of 'sS_name_CenpC_sel' and the first cell without spot selections for 'startMovie'.

You then have to make a 'full' spot selection (downstream functions won't work without this):

```
>> sS_name_CenpC = kitUpdateSpotSelections(sS_name_CenpC_sel, mS_name_paired,
[]);
```

Here, the empty brackets are for if you made your spot selection from a subset of cells, but you probably didn't! If you have had multiple pause/restarts on your spot selection, just put the most recent one in the place of sS_name_CenpC_sel (e.g. sS_name_CenpC_sel3). Save all sS structures you make (both sS_sel and sS).

5.3 Creating intensity and distance structures

5.3.1 Create intensity structure

This will extract intensity values from the movieStructure for the channels you select (can only be 2 channels at a time) only for the spots you selected. Points to consider:

- This defaults to accessing the intensity measurement in the vicinity of the 'inner' spot (with position corrected for chromatic shift for the secondary channel). We are assuming you have designated the reference kinetochore spot channel as the innermost channel back in Section 2.
    - If you aren't sure on what order your channels are in, open up your movieStructure within MATLAB, then navigate to cell 1 > options > neighbourSpots > channelOrientation. This will list your channels from inner to outer. You can also access this in the command window (no semicolon will permit the output to be printed to the command window):

      ```
      >> mS_name_paired{1}.options.neighbourSpots.channelOrientation
      ```

    - If you do not want the intensity to be measured in this way, change 'refMarker' to 'outer' for the intensity to be measured in the vicinity of the marker you designated as being 'outside' in Section 2, or 'self' for each intensity to be measured in the vicinity of the detected spot for their own channel (if detected, not recommended for variable kinetochore markers like Mad2).
    - If you have your 'outer' channel as the reference channel, that makes downstream graph plotting awkward as I based literally all of my plotting scripts on the reference kinetochore marker being ‘inner. Swap the inner and outer intensity measurements as shown in highlighted region below.
- Ideally you would normalise to your reference kinetochore marker, so 'channels' should include the reference kinetochore marker channel and the channel number of your intensity protein.

You just need a single intiM structure here:

```
>> intiM_exp01_paired_CenpCsel_CenpCin_Mad2out =
dublIntensityMeasurements(mS_name_paired, 'channels', [1 2], 'spotSelection',
sS_name_CenpC, 'paired', 1);
```

This intiM is named assuming that these proteins are indeed in this order – verify this as described above.

If you want to swap the orientation of your channels (heavily advised for downstream plotting if your reference channel is in the 'outer' position), please use the following function:

```
>> intiM_exp01_paired_CenpCsel_CenpCin_Mad2out =
kitSwapIntiMInnerOuter(intiM_exp01_paired_CenpCsel_CenpCout_Mad2in);
```

Save your intiM structure(s) in the same way as the movieStructure. You can drag these back into the workspace to reload them once you've closed MATLAB.

5.3.2 Create distance measurement structure

This creates an iM structure containing the inter-sister distance between your paired reference kinetochore marker spots. Note: You will also create iM structures later on where intra-kinetochore distances are needed (see section 6), but this forces the innermost channel to be the one that inter-sister distance is measured between. If that's fine for your data then great, if not then you will require the following function.

For 'channel', choose your reference kinetochore channel. This will create a stripped-back iM structure that is usable for pseudo-timeline plotting.

```
>> iM_name_CenpConly = dublSisSep(mS_name_paired, 'channel', 2, 'paired', 1,
'spotSelection', sS_name_CenpC);
```

5.4 Plotting inter-sister distances

The following function will split your kinetochores into bins based on their Mad2 intensity, then plot the median intensity of Mad2 (in black) and the median inter-sister distance per bin where the sister kinetochores are in the same or adjacent bins.

```
>> KKIntData = makeKKPlots(ctrl_iMintiMPairs, treat_iMintiMPairs)
```

ctrl_iMintiMPairs should contain iMs and intiMs from untreated, DMSO-treated, and/or wild-type cells (matched with treatIntiMs, if appropriate). They can be from a single experiment or multiple experiments. Note that if you wish to use an iM created from section 6.3.1, ensure that the innermost marker is the one you want to plot the inter-sister distance for. If not, create a new iM as in section 5.3.2. The iMs and intiMs should be in curly brackets, with the iM first and the Mad2 intiM second, separated by commas. If you have multiple experiments, each experiment should be separated by a semicolon, with commas in between the iM and intiM for a single experiment. For example:

```
{iM_exp01_paired_CenpConly, intiM_exp01_paired_CenpCsel_CenpCin_Mad2out;
iM_exp02_paired_CenpConly, intiM_exp02_paired_CenpCsel_CenpCin_Mad2out}
```

If you have multiple experiments (as shown here), there will be a first-pass ordering of kinetochores and normalising to 1 for each experiment; the normalised data from all experiments will then be combined, re-sorted and re-binned.

treat_iMintiMPairs should contain iMs and intiMs from drug-treated/perturbed cells, (e.g. knockout/ knockdown cell lines, or treated with nocodazole / MG132 / ZM / reversine / monastrol etc) matched with their corresponding ctrl_iMintiMPairs. **If you only have non-perturbed data, input this variable as [ ] (empty square brackets).** This function will normalise the intiMs for treatment to the intiMs of the control.

E.g. if your ctrl_iMintiMPairs consisted of:

```
{iM_exp01_DMSO_paired_CenpConly,
intiM_exp01_DMSO_paired_CenpCsel_CenpCin_Mad2out;
iM_exp02_DMSO_paired_CenpConly,
intiM_exp02_DMSO_paired_CenpCsel_CenpCin_Mad2out}
```

Your treat_iMintiMPairs should be ordered as such:

```
{iM_exp01_ZM_paired_CenpConly, intiM_exp01_ZM_paired_CenpCsel_CenpCin_Mad2out;
iM_exp02_ZM_paired_CenpConly, intiM_ZM_exp02_paired_CenpCsel_CenpCin_Mad2out}
```

This will normalise your treatment intensities to your control intensities (matched for each experiment), then split into the same number of Mad2 bins as the control, and then plot.

YOU CANNOT HAVE TREATINTIMS WITHOUT CTRLINTIMS !! I've never tried it but I'm 10000% sure attempting it will throw you an error.

There are options that you can add after treatIntiMs:

- 'nBins', 25 is default but you can change to any integer
- 'normIndInt' is 1 by default, but can change to 0. This normalises the Mad2 intensity to the reference kinetochore marker intensity (assuming these are arranged as 'outer' and 'inner', respectively)
- 'normalise' is 1 by default, but can change to 0. This scales all intensities such that the maximum median intensity for each marker is 1 for the control graph. I would advise keeping this as 1 if you have multiple experimental repeats.

Example: plotting one experiment without treatment data in 20 bins

```
>> exp01_KKbyMad2 = makeKKPlots({iM_exp01_DMSO_paired_CenpConly,
intiM_exp01_DMSO_paired_CenpCsel_CenpCin_Mad2out}, [], 'nBins', 20);
```

Example: plotting two experiments including treatment data in 25 bins

```
>> exp01_02_DMSO_ZM_KK = makeKKPlots{iM_exp01_DMSO_paired_CenpConly,
intiM_exp01_DMSO_paired_CenpCsel_CenpCin_Mad2out;
iM_exp02_DMSO_paired_CenpConly,
intiM_exp02_DMSO_paired_CenpCsel_CenpCin_Mad2out},
{iM_exp01_ZM_paired_CenpConly, intiM_exp01_ZM_paired_CenpCsel_CenpCin_Mad2out;
iM_exp02_ZM_paired_CenpConly, intiM_ZM_exp02_paired_CenpCsel_CenpCin_Mad2out});
```

The created variable KKIntData will allow you to access the normalised data and inter-sister (KK) distances that created the graph. If you have treatment and control datasets, the values for both of these can be accessed. The code will automatically calculate 'half change' for Mad2 and the inter-sister distance, i.e. the bins where the value of each of these have changed by half, and the intensity of Mad2 at the location where the inter-sister distance has changed by half. Save KKIntData to easily access the values used for plotting these graphs later on.

Your inter-sister and intensity curves will have three regions – a light shaded region corresponding to the lower and upper quartiles, a dark shaded region corresponding to the 95% confidence interval of the median, and a solid line corresponding to the median.

You can alter the dimensions of the plot relatively easily through the MATLAB command window (I recommend changing axis limits and dimensions in MATLAB as it’s a pain to do neatly in vector graphics software). I have made the code so that the treatment data axes are at the same scale as the control data – unfortunately, this might make the axes larger than can fit in your figure window/screen! If this happens, I recommend selecting your too-big figure, then typing the following in the MATLAB command window:

```
>> gca_pos = get(gca, 'Position')
```

This will print (in order) the left position, bottom position, width and height dimensions of your axes in pixels. You can then type:

```
>> get(0, 'ScreenSize')
```

This will give you your screen ‘position’ in pixels in the same order as the axes ‘position’ above. This will give you an idea of the relative scales that you are working with. Then you can change your axes ‘Position’ to work within that. You probably want to leave your left/bottom/width positions the same.

```
>> set(gca, 'Position', [gca_pos(1), gca_pos(2), gca_pos(3), NEW HEIGHT])
```

If you want to change the y-limits of the left (Mad2) or right (inter-sister distance) y-axis, then type the following, altered for what you want to change and what you want to change to:

```
>> yyaxis left
```

```
>> ylim([-0.05, 1.2])
```

This will keep the size of your axes the same, but will stretch/compress your data within that.

Once you are happy with your plot, in figure window, File > Save as, then save as .fig file. This will only open in MATLAB, but is best to access and use if you need to change axis scaling etc.

You can also save as .svg file for editing in vector programmes – you may have to change the figure renderer to 'painters' in MATLAB for this to be editable in your vector editing software (it may just export as a picture in .svg format otherwise). You can do this through the command window:

```
>> set(gcf, 'Renderer', 'painters')
```

6 Intra-kinetochore Delta distances

This is what we want when analysing Ndc80 to CenpC distance, for example. This corrects for overinflation in measurements.

6.1 Pairing

See section 5.1.

6.2 Spot selection

You may want to edit the kitSelectData_sS and kitSelectData_SelChan_sS files. Because of past MATLAB crashes that have made me lose hours of work, I have instructed the functions to save the spot selection as it goes in a folder, which is currently the folder you have open in the MATLAB panel. If that is suboptimal for you, e.g. if you have an analysis folder somewhere else you want to save to, then replace kitSelectData_sS line 93 and/or kitSelectData_SelChan_sS line 105 with the following:

```
filename = sprintf('FILEPATH TO DESIRED FOLDER%s%s_workingSpotSelection.mat',
folderSeparator, todayDate);
```

If this is giving you errors, make sure there are no extra forward/back slashes at the end of 'Filepath to desired folder'.

6.2.1 If you have already filtered reference kinetochore marker spots after pairing

If you have filtered the reference kinetochore marker spots post-pairing already in section 5.2, load in that sS_name_CenpC_sel structure, then proceed to 6.2.5.

6.2.2 If you have already selected good quality reference kinetochore marker spots for intensity measurements

If you have already screened your reference kinetochore marker spots for intensity measurements in 4.1.2, you can save some time by using the screened data to make spot selections for your new paired data via the following function:

```
>> sS_name_CenpC = kitGetPairedSpotSelFromFilteredData(mS_unpairedFiltered,
mS_pairedUnfiltered);
```

This will obtain the IDs of the acceptable spots from 4.1.2 and compare them to the spots you have accepted for pairing, and then remove any spots which did not pass your screening during 4.1.2. Save this sS structure, then proceed to 6.2.5.

6.2.3 If you have not done any spot filtering for this experiment at any point and you are only measuring intra-kinetochore Delta between one pair of markers

Do a spot selection for your kinetochore reference marker and secondary marker:

```
>> sS_name_CenpCOtherProtein_sel = kitSelectData_sS(mS_name_paired);
```

This will open up a window asking if you want to continue for each cell. Click 'yes' if you want to continue, 'skip' if you don't want to use this cell for some reason, or 'no' if you want to pause for a bit. Remember to save the sS structure after you've paused.

Look at the spot in your reference kinetochore marker channel and the channel of the marker you want to measure to (press the 'next chan' button to enable this), and if any of the following are true, remove:

- Spot centre is poorly detected (i.e. central cross is not in middle of spot)
- Spot is misshapen, i.e. is not Gaussian and symmetrical (mildly elliptical is okay)
- Spot is close to another spot (use intensity circle option in the bottom to check this, if the edge of the intensity circle is touching another spot, remove).
- Spot is very faint

Press finish after each cell. Once you've done all cells, you will get a spot selection structure.

Optional: If you're not done but you have to stop the selection for a substantial amount of time (e.g. when it's the end of the day, you have another experiment to do, etc), just press "no" when asking to continue spot selection. Then when you come back to it, take your old sS_name file and do 'addMore', sS_name as an option, and 'startMovie' as the next cell you need to start selecting from, e.g. if you completed selection up to cell 20 and now want to start from cell 21:

```
>> sS_name_CenpCOtherProtein_sel2 = kitSelectData_sS(mS_name_paired, 'addMore',
sS_name_CenpCOtherProtein_sel, 'startMovie', 21);
```

If you have to pause multiple times, then change the variable names to reflect how many times you've had to pause, and make sure your startMovie number is correct, e.g.:

```
>> sS_name_CenpCOtherProtein_= kitSelectData_sS(mS_name_paired, 'addMore',
sS_name_CenpCOtherProtein_sel2, 'startMovie', 42);
```

Increment each sel number by 1 each time you re-start for the same experiment's spot selection. Save the newest sS structure every time you pause.

If MATLAB crashes, navigate to the folder that contains your 'workingSpotSelection', and drag it back into MATLAB. If you are unsure which cell you have got to, click into that spot selection, then navigate to the data-containing portion of the 'selection' substructure and scroll to the bottom. The number at the bottom of the left-hand column is the last cell that you performed spot selections for. You can then continue spot selections as described in the highlighted blue section above, using the workingspotselection structure as the selection for 'addMore' in place of 'sS_name_CenpCOtherProtein_sel' and the first cell without spot selections for 'startMovie'.

You then have to make a 'full' spot selection (downstream functions won't work without this):

```
>> sS_name_CenpCOtherProtein =
kitUpdateSpotSelections(sS_name_CenpCOtherProtein_sel, mS_name_paired, []);
```

Here, the empty brackets are for if you made your spot selection from a subset of cells, but you probably didn't! If you have had multiple pause/restarts on your spot selection, just put the most recent one in the place of sS_name_CenpCOtherProtein_sel (e.g. sS_name_CenpCOtherProtein_sel3).

Save all sS structures you make (both sS_sel and sS).

6.2.4 If you have not done any spot filtering for this experiment at any point and you are measuring intra-kinetochore Delta between more than one pair of markers

Do a first round of spot selection for the reference kinetochore marker channel only as described in 5.2.2. Then proceed to 6.2.5.

6.2.5 If you have a pre-existing spotSelection structure with spot filtering for the reference kinetochore marker channel only

Single distance measurement

Load in the reference channel only spot selection file. This can be any type of spot selection (either with or without _sel at the end). Then run another round of spot selection for the channel that you want.

```
>> sS_name_CenpCOtherProtein_sel = kitSelectData_SelChan_sS(mS_name_paired,
'prevSel', sS_name_CenpC_sel, 'channel', 1/2/3);
```

This will open up a window asking if you want to continue for each cell. Click 'yes' if you want to continue, 'skip' if you don't want to use this cell for some reason, or 'no' if you want to pause for a bit. Remember to save the sS structure after you've paused.

Look at the spot in your desired channel (don't click 'next chan' as you will then not be filtering in the channel you care about), and if any of the following are true, remove:

- Spot centre is poorly detected (i.e. central cross is not in middle of spot)
- Spot is misshapen, i.e. is not Gaussian and symmetrical (mildly elliptical is okay)
- Spot is close to another spot (use <u>intensity circle</u> option in the bottom to check this, if the edge of the intensity circle is touching another spot, remove).
- Spot is very faint

Press <u>finish</u> after each cell. Once you've done all cells, you will get a spot selection structure.

Optional: If you're not done but you have to stop the selection for a substantial amount of time (e.g. when it's the end of the day, you have another experiment to do, etc), just press "<u>no</u>" when asking to continue spot selection. Then when you come back to it, take your old sS_name file and do 'addMore', sS_name as an option, and 'startMovie' as the next cell you need to start selecting from, e.g. if you completed selection up to cell 20 and now want to start from cell 21:

```
>> sS_name_CenpCOtherProtein_sel2 = kitSelectData_SelChan_sS(mS_name_paired,
'prevSel', sS_name_CenpC_sel, 'channel', 1/2/3, 'addMore',
sS_name_CenpCOtherProtein_sel, 'startMovie', 21);
```

If you have to pause multiple times, then change the variable names to reflect how many times you've had to pause, and make sure your startMovie number is correct, e.g.:

```
>> sS_name_CenpCOtherProtein_sel3 = kitSelectData_SelChan_sS(mS_name_paired,
'prevSel', sS_name_CenpC_sel, 'channel', 1/2/3, 'addMore',
sS_name_CenpCOtherProtein_sel2, 'startMovie', 42);
```

Increment each sel number by 1 each time you re-start for the same experiment's spot selection. Save the newest sS structure every time you pause.

If MATLAB crashes, navigate to the folder that contains your 'workingSpotSelection', and drag it back into MATLAB. If you are unsure which cell you have got to, click into that spot selection, then navigate to the data-containing portion of the 'selection' substructure and scroll to the bottom. The number at the bottom of the left-hand column is the last cell that you performed spot selections for. You can then continue spot selections as described in the highlighted blue section above, using the workingspotselection structure as the selection for 'addMore' in place of 'sS_name_CenpCOtherProtein_sel' and the first cell without spot selections for 'startMovie'.

You then have to make a 'full' spot selection (downstream functions won't work without this):

```
>> sS_name_CenpCOtherProtein =
kitUpdateSpotSelections(sS_name_CenpCOtherProtein_sel, mS_name_paired, []);
```

Here, the empty brackets are for if you made your spot selection from a subset of cells, but you probably didn't! If you have had multiple pause/restarts on your spot selection, just put the most recent one in the place of sS_name_CenpCOtherProtein_sel (e.g. sS_name_CenpCOtherProtein_sel3).

Save all sS structures you make (both sS_sel and sS).

Pair-wise distance comparisons

If you want to do pair-wise distance comparisons between all three markers, then run two fresh sets of spot selection without looking at the reference kinetochore marker:

```
>> sS_name_SecondProtein_sel = kitSelectData_SelChan_sS(mS_name_paired,
'channel', 1/2/3);

>> sS_name_ThirdProtein_sel = kitSelectData_SelChan_sS(mS_name_paired,
'channel', 1/2/3);
```

This will open up a window asking if you want to continue for each cell. Click 'yes' if you want to continue, 'skip' if you don't want to use this cell for some reason, or 'no' if you want to pause for a bit. Remember to save your sS structure if you're pausing.

Look at the spot, and if any of the following are true, remove:

- Spot centre is poorly detected (i.e. central cross is not in middle of spot)
- Spot is misshapen, i.e. is not Gaussian and symmetrical (mildly elliptical is okay)
- Spot is close to another spot (use intensity circle option in the bottom to check this, if the edge of the intensity circle is touching another spot, remove).
- Spot is very faint

Don't click 'next chan' as that will result in you filtering in more than the desired kinetochore marker channel.

Press finish after each cell. Once you've done all cells, you will get a spot selection structure.

Optional: If you're not done but you have to stop the selection for a substantial amount of time (generally more than 2h, e.g. when it's the end of the day, you have another experiment to do, etc), just press "no" when asking to continue spot selection. Make sure you save the spot selection so far. Then when you come back to it, take your old sS_name file and do 'addMore', sS_name as an option, and 'startMovie' as the next cell you need to start selecting from, e.g. if you completed selection up to cell 20 and now want to start from cell 21:

```
>> sS_name_SecondProtein_sel2 = kitSelectData_SelChan_sS(mS_name_paired,
'channel', 1/2/3, 'addMore', sS_name_SecondProtein_sel, 'startMovie', 21);
```

If you have to pause multiple times, then change the variable names to reflect how many times you've had to pause, and make sure your startMovie number is correct, e.g.:

```
>> sS_name_SecondProtein_sel3 = kitSelectData_SelChan_sS(mS_name_paired,
'channel', 1/2/3, 'addMore', sS_name_SecondProtein_sel2, 'startMovie', 42);
```

Increment each sel number by 1 each time you re-start for the same experiment's spot selection. Save your sS structure every time you pause.

If MATLAB crashes, navigate to the folder that contains your 'workingSpotSelection', and drag it back into MATLAB. If you are unsure which cell you have got to, click into that spot selection, then navigate to the data-containing portion of the 'selection' substructure and scroll to the bottom. The number at the bottom of the left-hand column is the last cell that you performed spot selections for. You can then continue spot selections as described in the highlighted blue section above, using the workingspotselection structure as the selection for 'addMore' in place of 'sS_name_SecondProtein_sel' and the first cell without spot selections for 'startMovie'.

Then you can combine the spot selections for the appropriate pair wise measurements. If you have had multiple pause/restarts on your spot selection, just put the most recent one in the place of sS_name_{proteinSelections}_sel, e.g. sS_name_{proteinSelections}_sel3:

```
>> sS_name_CenpCSecondProtein_sel =
kitCombineChannelSpotSelections(sS_name_CenpC_sel, sS_name_SecondProtein_sel);

>> sS_name_CenpCThirdProtein_sel =
kitCombineChannelSpotSelections(sS_name_CenpC_sel, sS_name_ThirdProtein_sel);
```

```
>> sS_name_SecondProteinThirdProtein_sel =
kitCombineChannelSpotSelections(sS_name_SecondProtein_sel,
sS_name_ThirdProtein_sel);
```

You then have to make a 'full' spot selection (downstream functions won't work without this):

```
>> sS_name_CenpCSecondProtein =
kitUpdateSpotSelections(sS_name_CenpCSecondProtein_sel, mS_name_paired, []);

>> sS_name_CenpCThirdProtein =
kitUpdateSpotSelections(sS_name_CenpCThirdProtein_sel, mS_name_paired, []);

>> sS_name_SecondProteinThirdProtein =
kitUpdateSpotSelections(sS_name_SecondProteinThirdProtein_sel, mS_name_paired,
[]);
```

Here, the empty brackets are for if you made your spot selection from a subset of cells, but you probably didn't!

Save all sS structures you make (both sS_sel and sS).

6.3 Obtaining binned intra-kinetochore Delta

6.3.1 Perform 3D intra-kinetochore measurements

This creates and compiles 1D (in x, y and z), 2D (xy only), and 3D distance measurements between the spots of your two markers where both markers have passed filtering, finely corrected for per-cell chromatic aberrations. 'channels' should be the channel numbers of your reference kinetochore marker and the channel number of your other protein, I have just written [2 3] below for ease.

```
>> iM_name_CenpCOtherProtein = dublIntraMeasurements(mS_name_paired, 'paired',
1, 'channels', [2 3], 'spotSelection', sS_name_CenpCOtherProtein);
```

Save this structure.

6.3.2 Measure Mad2 intensity

This will extract intensity values from the movieStructure for the channels you select (can only be 2 channels at a time) only for the spots you selected. Points to consider:

- This defaults to accessing the intensity measurement in the vicinity of the 'inner' spot (with position corrected for chromatic shift for the secondary channel). We are assuming you have designated the reference kinetochore spot channel as the innermost channel back in Section 2.
    - If you aren't sure on what order your channels are in, open up your movieStructure within MATLAB, then navigate to cell 1 > options > neighbourSpots > channelOrientation. This will list your channels from inner to outer. You can also access this in the command window (no semicolon will permit the output to be printed to the command window):

      ```
      >> mS_name_paired{1}.options.neighbourSpots.channelOrientation
      ```

    - If you do not want the intensity to be measured in this way, change 'refMarker' to 'outer' for the intensity to be measured in the vicinity of the marker you designated as being 'outside' in Section 2, or 'self' for each intensity to be measured in the vicinity of the detected spot for their own channel (if detected, not recommended in the case of variable epitopes like Mad2).
    - If you have your 'outer' channel as the reference channel, that makes downstream graph plotting awkward as I based literally all of my plotting scripts on the reference kinetochore marker being ‘inner’. Swap as shown in highlighted region.
- Ideally you would normalise to your reference kinetochore marker, so 'channels' should include the reference kinetochore marker channel and the channel number of your intensity protein.
- You should tell the intensity measurement function that the data you are providing is paired, this is so it collates the intensities in a structure comparable to the iM.

```
>> intiM_exp01_paired_CenpCOtherProteinsel_CenpCin_Mad2out =
dublIntensityMeasurements(mS_name_paired, 'channels', [1 2], 'spotSelection',
sS_name_CenpCOtherProtein, 'paired', 1);
```

The reason that we are applying the spot selection for CenpC and the non-Mad2 protein is that we are almost never measuring the intra-kinetochore delta to Mad2, we are just finding the Mad2 intensity of the kinetochores with good enough spot centre localisation for the two markers we are measuring between. This intensity will then be utilised for splitting the kinetochores which satisfied the pass criteria into bins.

This intiM is named assuming that these proteins are indeed in this order – verify this as described above.

If you want to swap the orientation of your channels (heavily advised for downstream plotting if your reference channel is in the 'outer' position), please use the following function:

```
>> intiM_exp01_paired_CenpCOtherProteinsel_CenpCin_Mad2out =
kitSwapIntiMInnerOuter(intiM_exp01_paired_CenpCOtherProteinsel_CenpCout_Mad2in)
;
```

Save your intiM structure.

6.3.3 Split data into bins based on Mad2 intensity

This will split your kinetochores into bins that will then get run through the overinflation correction pipeline.

Optional: combine data from multiple experiments

You may have multiple experiments that you are running this analysis on. You should run BEDCA on the individual experiments first. This will allow you to ensure that the individual experiments are sufficiently similar to warrant combining. This function gives you a combined iM (distance measurements) and intiM (intensity measurements) for your multiple experiments. The data contained in the ‘outer’ marker of the intiM (generally Mad2) will default to being normalised to the ‘inner’ marker (CenpC for us) and then normalised either to 1 or such that the control intensities would be equal to 1. This normalised ‘outerDivInner’ intensity will be stored in the ‘outer’ portion of the intiM structure.

```
>> [iM_expNums_CenpCOtherProtein, intiM_expNums_Mad2divCenpC] =
kitNormCombineiMsIntiMs(iMs, intiMs, ctrlintiMs);
```

For control or untreated data, **ctrlintiMs** can be input as empty square brackets (**[ ]**).

For treatment data or perturbed cell lines, **ctrlintiMs** should be the Mad2 intiMs made by the control dataset (i.e. DMSO treated or WT cells).

iMs, intiMs and ctrlIntiMs should be input in curly brackets, with the data corresponding to each experiment in the same position within each, i.e. if you have experiments 3 and 4, treated with nocodazole vs those treated with DMSO:

iMs should be:

```
{iM_exp03_noc_CenpCNdc80, iM_exp04_noc_CenpCNdc80}
```

intiMs should be:

```
{intiM_exp03_noc_paired_CenpCNdc80sel_CenpCin_Mad2out,
intiM_exp04_noc_paired_CenpCNdc80sel_CenpCin_Mad2out}
```

ctrlIntiMs should be:

```
{intiM_exp03_DMSO_paired_CenpCNdc80sel_CenpCin_Mad2out,
intiM_exp04_DMSO_paired_CenpCNdc80sel_CenpCin_Mad2out}
```

For non-treated data, you will type something like:

```
>> [iM_exp03_04_CenpCNdc80,
intiM_exp03_04_paired_CenpCNdc80sel_Mad2divCenpCout] =
kitNormCombineiMsIntiMs({iM_exp03_CenpCNdc80, iM_exp04_CenpCNdc80},
{intiM_exp03_paired_CenpCNdc80sel_CenpCin_Mad2out,
intiM_exp04_paired_CenpCNdc80sel_CenpCin_Mad2out}, [], 'nBins', 25);
```

For treatment data, you will type something like:

```
>> [iM_exp03_04_noc_CenpCNdc80,
intiM_exp03_04_noc_paired_CenpCNdc80sel_Mad2divCenpCout] =
kitNormCombineiMsIntiMs({iM_exp03_noc_CenpCNdc80N, iM_exp04_noc_CenpCNdc80N},
{intiM_exp03_noc_paired_CenpCNdc80sel_CenpCin_Mad2out,
intiM_exp04_noc_paired_CenpCNdc80sel_CenpCin_Mad2out},
{intiM_exp03_DMSO_paired_CenpCNdc80sel_CenpCin_Mad2out,
intiM_exp04_DMSO_paired_CenpCNdc80sel_CenpCin_Mad2out}, 'nBins', 25);
```

The data is automatically set to be split into 25 bins before combining, you can change this by changing 25 to another integer. If you didn’t want to normalise Mad2 to CenpC, I would add ‘normalisationMarker’, 'outer' at the end (before the closing brackets) – this will then just combine and normalise based on Mad2 intensity. There is no option to not normalise to 1, and that is intentional!

Split the data

Navigate to the KiT-master-2.4.1 / BEDCA_v1 / BEDCA / ExptData folder within MATLAB – make sure this is the folder which is selected when reading the file path at the top of the command window. This is because the next function will save a variable in the current folder, and we need the created variable to save within ExptData.

Now you can split data into bins:

```
>> makeiMinSteps(nBins, IntiM, iM, normalised, markerdist, intID, expnum,
expcond)
```

Say I want to measure the distance between CenpC and Ndc80 and split this data into 25 bins based on Mad2/CenpC intensity, I am on experiment 03, and have done a DMSO treatment. I would write the following:

```
>> makeiMinSteps(25, intiM_exp03_DMSO_Mad2CenpCNdc80N_CenpCin_Mad2out,
iM_exp03_DMSO_Mad2CenpCNdc80_CenpCNdc80, 'normalised', 'CenpCNdc80N', 'Mad2',
'03', 'DMSO')
```

This will automatically make and save an iM of iMs in the folder you've navigated to (ExptData). The main iM name will be something like this:

```
iM_exp03_distCenpCNdc80_NintMad2_ss168_DMSO
```

Whilst the binned iM names (contained within the main name) will look like this:

```
iM_exp03_distCenpCNdc80_NintMad2_ss168_DMSOBin01
```

If I didn't want to normalise Mad2 intensity to CenpC, I put empty quote marks in place of 'normalised' which would result in the final name not having N in front of intMad2.

If my cells were untreated I would put the expcond variable as empty quotes.

If I combined multiple experiments, I would write the following. Note that I have removed 'normalised' and I have replaced 'Mad2' with 'NormMad2' to reflect that I have normalised Mad2 to CenpC in the experiment combining pipeline. If you split your data into a different number than 25 bins when combining, then you should change 25 to the appropriate integer:

```
>> makeiMinSteps(25, intiM_exp03_04_paired_CenpCNdc80sel_Mad2divCenpCout,
iM_exp03_04_CenpCNdc80, '', 'CenpCNdc80', 'NormMad2', '0304', '')
```

If you accidentally saved your split iM structure in another folder, make sure you move it to your ExptData folder.

6.4 Derive intra-kinetochore Delta per bin

Update ExptData_list.m file:

Navigate up one level to the KiT-master-2.4.1 / BEDCA_v1 / BEDCA folder.

Open the ExptData_list.m file.

You can choose to change the ExptDataLoc variable to something like this:

```
ExptDataLoc = 'FULL PATH TO MATLAB FOLDER/MATLAB/KiT-master-
2.4.1/BEDCA_v1/BEDCA/ExptData/';
```

This is to stop MATLAB from erroring if you're accidentally in the wrong folder when you start running BEDCA. Provided you're in the KiT-master-2.4.1 / BEDCA_v1 / BEDCA folder, you should have no problems if you leave this variable unchanged. But I have accidentally run BEDCA when I'm in the "wrong" folder too many times so changed it for myself!

Then change the FileNames{1} variable:

```
FileNames{1} = 'iM_exp03_distCenpCNdc80_NintMad2_ss168_DMSO.mat';
```

This should be the name of the file which contains substructures with your binned data, it should not have 'Bin01' or similar in its filename as BEDCA will get confused. Assuming you have used the makeiMinSteps function, this shouldn't be a problem.

Then you will need to change/add the associated ExptDats.name variables:

```
FileNames{1}.name = 'iM_exp03_distCenpCNdc80_NintMad2_ss168_DMSOBin01.mat';
FileNames{2}.name = 'iM_exp03_distCenpCNdc80_NintMad2_ss168_DMSOBin02.mat';
.
.
.
FileNames{25}.name = 'iM_exp03_distCenpCNdc80_NintMad2_ss168_DMSOBin25.mat';
```

Then you may need to change the associated ExptDats.treatment variables:

```
ExptDats{1}.treatment = 'Bin01';
ExptDats{2}.treatment = 'Bin02';
.
.
.
ExptDats{25}.treatment = 'Bin25';
```

Then you will need to change the associated ExptDats.fluorophore variables. If you have used ONLY antibodies for the markers that you will be measuring Delta between then this should be 'Ab', otherwise if you are measuring distances between antibody and fluorescent proteins then you should use 'GFP' because fluorescent proteins' spots are generally a bit less nice than antibodies so will use different priors.

```
ExptDats{1}.fluorophore  = 'Ab';
ExptDats{2}.fluorophore  = 'Ab';
.
.
.
ExptDats{25}.fluorophore  = 'Ab';
```

Update treatment library

Scroll up to TreatmentLib. You may need to add Bin01 etc to this:

```
TreatmentLib = {{'DMSO','DMSO'}, {'Bin01','Bin01'}, {'Bin02','Bin02'}, ...
{'Bin25','Bin25'}};
```

Command + S to save the ExptData_list.m file.

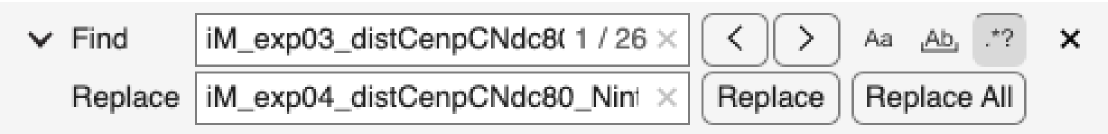
Pro tip: if you press command + F within the ExptData_list.m file and then use the little drop-down button to do find and replace, you can do this bit in about 30 seconds. Make 'Find' the name of FileNames{1}, excluding the quotes and .mat, and 'Replace' the file name of your new structure of binned iMs excluding .mat, e.g. Find:

```
iM_exp03_distCenpCNdc80_NintMad2_ss168_DMSO
```

Replace:

```
iM_exp04_distCenpCNdc80_NintMad2_ss174_DMSO
```

Then click 'Replace All'.



Run BEDCA

BEDCA stands for Bayesian Euclidean Distance Correction Algorithm. We use it to correct for overinflation bias arising from measurement noise when measuring small 3D distances. See main and supplementary figures of https://doi.org/10.1016/j.celrep.2020.107535.

This pipeline uses exactly the same version of BEDCA as utilised in 'Ensemble-Level Organization of Human Kinetochores and Evidence for Distinct Tension and Attachment Sensors' (https://doi.org/10.1016/j.celrep.2020.107535) and 'Subcellular Euclidean distance measurements with multicolor fluorescence localization imaging in cultured cells' (https://doi.org/10.1016/j.xpro.2021.100774), with the BEDCA_v1 code within the KiT-master-2.4.1 folder taken directly from https://github.com/cmcb-warwick/KiD-also-known-as-KiT2.1.10-and-BEDCA.

In the command line, type the following. Note, if you fail to converge then increase nsteps (I tend to double each time until I get to 1.6 million, at which point if it hasn't converged then it never will). See also steps 13-15 of https://doi.org/10.1016/j.xpro.2021.100774.

```
>> Run_BEDCA(1, 'Bin01', [], [], struct('nsteps', 100000));
```

Make sure the run has converged (GRc statistic < 1.02, it will show up in the command window if this hasn't happened – the GRc statistic will also be in the figures that pop up at the end of the run). If it has, then navigate to MATLAB / KiT-master-2.4.1 / BEDCA_v1 / MCMCruns in your file explorer, go to the right folder for your experiment, and open 'SummaryStats_chain1.csv'. This will tell you where the data has converged to. It has the mean, median, and standard deviation for:

- The 'mu' of the distribution (the approximate mean of the distribution – this is the number you will use most often!)
- The 'taux' and 'tauz' values (the precision in xy and z, respectively. Generally taux > tauz)

- The 'sdx' and 'sdx' values (the estimated prior values used for convergence for xy and z).

Once finished, change 'Bin01' to 'Bin02' etc and re-run until you have run all your data.

This will take up **many** gigabytes of data on your computer (at least 20GB for 80 cells).

6.5 Creating Delta plots

6.5.1 Create variables for Mean and Standard Deviation of Mu (Delta)

Create a new MATLAB script window to collate your Mean and Standard Deviations from your Mu statistic. This is the mean and standard deviation of the Monte Carlo Markov Chain samples from the posterior distribution estimating the true mean of the sample. For each bin, navigate to MATLAB / KiT-master-2.4.1 / BEDCA_v1 / MCMCruns / [Appropriate experiment/bin folder] and open 'SummaryStats_chain1.csv'. The first row gives you the mean, median, and standard deviation for Mu.

Navigate to the 'Editor' tab on the top of the MATLAB window, then click 'New' in the top left of this tab. This will open up a new MATLAB script that you can edit.

In your MATLAB script window, create two new variables:

```
exp03_meanMu = [ ];

exp03_stdDevMu = [ ];
```

Then copy and paste your bin 1 mean and standard deviation for the mu statistic into the relevant variable:

```
exp03_meanMu = [14.85848028,  ];

exp03_stdDevMu = [8.503912284, ];
```

Do this for all 25 bins (or however many bins you did). Then run the script by pressing command + Enter. This will put the variables into the workspace. You can also save this script to permit you to access the values held there for the future. You can also save the variables in the workspace as previously described for intiMs and iMs.

6.5.2 Make the plot

The following function will split your kinetochores into bins based on their Mad2 intensity, then plot the intensity of Mad2 (in black) and your intra-kinetochore Delta values (in pink) within those bins.

```
>> deltaData = makeDeltaPlots(deltaIntiM, normIntiM)
```

deltaIntiM must be organised as {delta, stddev, Mad2intiM}. This function can only handle one set of data, as averaging the mean Mu from multiple experiments is mathematically invalid for producing a combined Mu – you will have to rerun BEDCA with combined experimental data.

normIntiM should only be provided if you have only run BEDCA on a single experiment. Otherwise, input this as empty square brackets (**[ ]**).

There are options:

- normalise: 0 or {1}. Whether to normalise the median of the maximum bin for each intensity class to one. Will normalise to normData or intiM you have provided. **If you are plotting a treatment dataset combined from multiple experiments, make this 0.**
- normIndInt: {0} or 1. Whether to normalise the independent intensity marker (i.e. Mad2) to CenpC or not. Because the pooled data has been binned on Mad2/CenpC intensity, this is generally 0. **If you are plotting from a single experiment (untreated or treatment), change this to 1.**
- normIndInt_ctrl: 0 or {1}. Whether to normalise the independent intensity marker of the control data you've provided (i.e. Mad2) to CenpC or not.
- intMarker: {nan} or string of intensity marker that you binned on. Will request this later if not filled in now.
- distPair: {nan} or string of proteins you are measuring delta between. Will request this later if not filled in now.

Example: you have run BEDCA for an untreated dataset for a single experiment

```
>> deltaData = makeDeltaPlots({exp03_meanMu, exp03_stdMu,
intiM_exp03_paired_CenpCNdc80sel_CenpCin_Mad2out}, [], 'normIndInt', 1);
```

Example: you have run BEDCA for a treated dataset for a single experiment

```
>> deltaData = makeDeltaPlots({exp03_noc_meanMu, exp03_noc_stdMu,
intiM_exp03_noc_paired_CenpCNdc80sel_CenpCin_Mad2out},
{intiM_exp03_DMSO_paired_CenpCNdc80sel_CenpCin_Mad2out}, 'normIndInt', 1);
```

Example: you have run BEDCA for an untreated dataset for multiple experiments

```
>> deltaData = makeDeltaPlots({exp03_04_meanMu, exp03_04_stdMu,
intiM_exp03_04_paired_CenpCNdc80sel_Mad2divCenpCout}, [], 'normIndInt', 0);
```

Example: you have run BEDCA for a treated dataset for a single experiment

```
>> deltaData = makeDeltaPlots({exp03_04_noc_meanMu, exp03_04_noc_stdMu,
intiM_exp03_04_noc_paired_CenpCNdc80sel_Mad2divCenpCout}, [], 'normIndInt', 0,
'normalise', 0);
```

The created variable deltaData will allow you to access the intensity and intra-kinetochore Delta values that created the graph. This doesn't create a separate subsection for treatment and control datasets, so you will have to keep track of this! The code will automatically calculate 'half change' for Mad2 and the intra-kinetochore Delta, i.e. the bin where the value of each have changed by half, and the intensity of Mad2 at the location where the intra-kinetochore Delta has changed by half. Save deltaData to easily access the values used for plotting these graphs later on.

Your intensity curve will have three regions – a light shaded region corresponding to the lower and upper quartiles, a dark shaded region corresponding to the 95% confidence interval of the median, and a solid line corresponding to the median. Your Delta curve will have two regions – a shaded region corresponding to the 95% confidence interval of the mean Mu (i.e. 1.96× the standard deviation), and a solid line corresponding to the mean Mu.

You can alter the dimensions of the plot relatively easily through the MATLAB command window (I recommend changing axis limits and dimensions in MATLAB as it's a pain to do neatly in vector graphics software). I have made the code so that the treatment data axes are at the same scale as the control data – unfortunately, this might make the axes larger than can fit in your figure window/screen! If this happens, I recommend selecting your too-big figure, then typing the following in the MATLAB command window:

```
>> gca_pos = get(gca, 'Position')
```

This will print (in order) the left position, bottom position, width and height dimensions of your axes in pixels. You can then type:

```
>> get(0, 'ScreenSize')
```

This will give you your screen 'position' in pixels in the same order as the axes 'position' above. This will give you an idea of the relative scales that you are working with. Then you can change your axes 'Position' to work within that. You probably want to leave your left/bottom/width positions the same.

```
>> set(gca, 'Position', [gca_pos(1), gca_pos(2), gca_pos(3), NEW HEIGHT])
```

If you want to change the y-limits of the left (Mad2) or right (Delta) y-axis, then type the following, altered for what you want to change and what you want to change to:

```
>> yyaxis left
```

```
>> ylim([-0.05, 1.2])
```

This will keep the size of your axes the same, but will stretch/compress your data within that.

Once you are happy with your plot, in figure window, File > Save as, then save as .fig file. This will only open in MATLAB, but is best to access and use if you need to change axis scaling etc.

You can also save as .svg file for editing in vector programmes – you may have to change the figure renderer to 'painters' in MATLAB for this to be editable in your vector editing software (it may just export as a picture in .svg format otherwise). You can do this through the command window:

```
>> set(gcf, 'Renderer', 'painters')
```

7 Performing intra-kinetochore Delta alongside inter-sister and/or three-channel intensity measurements on the same kinetochores

7.1 Pair kinetochores

See section 5.1 for details.

7.2 Spot selection

See section 6.2.3 for details.

7.3 Obtaining binned intra-kinetochore Delta

See section 6.3 for details.

7.4 Derive intra-kinetochore Delta per bin

See section 6.4 for details.

7.5 Plot intra-kinetochore Delta per bin

See section 6.5 for details.

7.6 Plot inter-sister distance per bin (optional)

Largely the same as 5.4, except the iMs and intiMs used should generally be those created as in section 6.3.1 and 6.3.2, respectively (i.e. using the spot selection in 2 channels). The only exception to this is if you want to measure inter-sister distance from a specified kinetochore marker that is not the innermost of the distance pair, then you would make and use the following iM (see also 5.3.2):

```
>> iM_name_CenpCSecondProteinsel = dublSisSep(mS_name_paired, 'channel', 2,
'paired', 1, 'spotSelection', sS_name_CenpCOtherProtein);
```

7.7 Obtain three-channel intensity measurements (optional)

Largely the same as 6.3.2, but you should create two separate intensity structures as below:

```
>> intiM_exp01_paired_CenpCOtherProteinsel_CenpCin_Mad2out =
dublIntensityMeasurements(mS_name_paired, 'channels', [1 2], 'spotSelection',
sS_name_CenpCOtherProtein, 'paired', 1);

>> intiM_exp01_paired_CenpCOtherProteinsel_CenpCin_OtherProteinout =
dublIntensityMeasurements(mS_name_paired, 'channels', [2 3], 'spotSelection',
sS_name_CenpCOtherProtein, 'paired', 1);
```

This is because you will probably want to measure the intensity of the proteins in the same (chromatic shift-corrected) location as one another. This is most reliable when you are using the spots detected in the reference kinetochore marker channel, as they should be present throughout mitosis and not subject to substantial changes that may affect their detection.

7.8 Plotting intensity measurement pseudo-timeline graphs (optional)

Largely as section 4.3, except you should ensure you use the intiMs created using the same spot selection as used in the intra-kinetochore Delta measurements, e.g.:

```
{intiM_exp01_paired_CenpCOtherProteinsel_CenpCin_Mad2out,
intiM_exp01_paired_CenpCOtherProteinsel_CenpCin_OtherProteinout}
```

8 Performing inter-sister and three-channel intensity measurements on the same kinetochores

8.1 Pair kinetochores

See section 5.1 for details.

8.2 Spot selection

See section 5.2.2 for details.

8.3 Creating intensity and distance structures

Largely the same as 5.3, but you should create two separate intensity structures as below following the guidance from section 5.3.1:

```
>> intiM_exp01_paired_CenpCsel_CenpCin_Mad2out =
dublIntensityMeasurements(mS_name_paired, 'channels', [1 2], 'spotSelection',
sS_name_CenpC, 'paired', 1);
```

```
>> intiM_exp01_paired_CenpCsel_CenpCin_SKAPout =
dublIntensityMeasurements(mS_name_paired, 'channels', [2 3], 'spotSelection',
sS_name_CenpC, 'paired', 1);
```

This is because you will probably want to measure the intensity of the proteins in the same (chromatic shift-corrected) location as one another. This is most reliable when you are using the spots detected in the reference kinetochore marker channel, as they should be present throughout mitosis and not subject to substantial changes that may affect their detection.

8.4 Plot inter-sister distance per bin

See section 5.4 for details.

8.5 Plotting intensity measurement pseudo-timeline graphs

Largely as section 4.3, except you should ensure you use the intiMs created in section 8.3 e.g.:

```
{intiM_exp01_paired_CenpCsel_CenpCin_Mad2out,
intiM_exp01_paired_CenpCsel_CenpCin_OtherProteinout}
```